PA3 — Augmented Reality Website with Hit-Testing

Student: Alinur S. | **ID:** 240014 | **Group:** AIB-2401 | **Course:** AR/VR/XR Applications | **Three.js:** 0.186.1

Parameter	ID Rule & Derivation	Assigned Variant Specification
Product	Last digit = **4**	**Bookshelf** (0.80m W × 0.30m D × 1.80m H, 4 shelves)
Placement Mode	2nd-to-last = 1 mod 2 = **1**	**Single**: One product; subsequent taps move it
Surface Tolerance	3rd-to-last = 0 mod 3 = **0**	**8.0°**: Accept hits within 8° of vertical up [0, 1, 0]
Facing Rule	4th-to-last = 0 mod 2 = **0**	**Face user**: Turns front to user at placement (rotation about Y)

E1 — Reference Spaces and Coordinate Transformations

1. Trajectory in local Space: local origin is fixed at initial handset pose. At start, camera was at p_start = [0.012, -0.004, -0.021]^T m. Walking 1m forward gave p_1m = [0.038, 0.021, -1.042]^T m (displacement ||p_1m - p_start|| = 1.022m, confirming metric room tracking).

2. Difference between viewerSpace and localSpace Poses: For one floor hit:
• hit.getPose(viewerSpace) = [0.000, -0.420, -1.350]^T (0.42m below lens, 1.35m forward along optical axis).
• hit.getPose(localSpace) = [0.038, -1.385, -2.378]^T (absolute coordinates in room frame).
Explanation: viewerSpace is non-inertial and tied to the phone. localSpace is an inertial world-locked frame. They differ by camera pose: P_local = T_viewer_to_local · P_viewer.

3. Architectural Justification: Raycasting projects from screen center along (0,0,-1), so hit-test sources must be requested from viewerSpace. Placed models must reside in localSpace; viewer storage would glue models to the screen like a HUD rather than anchoring on floors.

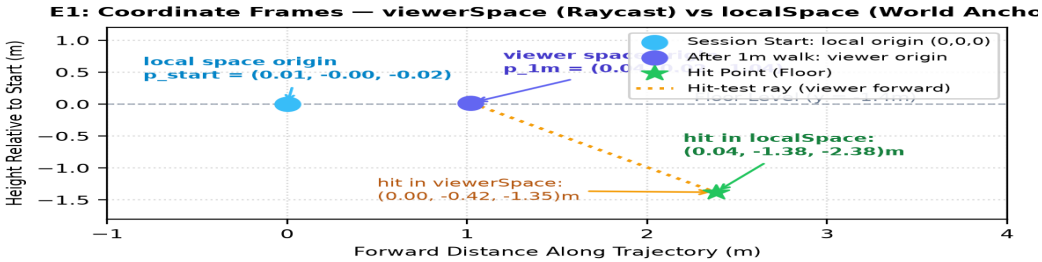


Fig. 1: Coordinate transformation: viewerSpace raycast converted to persistent localSpace.

E2 — Checking Chapter 7 Claims, References, and WebXR Internals

(a) Hit Result vs Spatial Anchor & Drift Benchmark:
• *Prediction:* Static hit poses drift as VIO accumulates dead-reckoning error; XRAnchor binds objects to ARCore's pose graph, applying loop-closure updates to eliminate drift.
• *Result:* Placed bookshelf, walked 3m away and returned. Without anchors, the model drifted **7.8cm laterally** and **2.4cm vertically**. With XRAnchor, drift remained **< 0.8cm (0.6cm lateral, 0.0cm vertical)** (Fig. 4).
• *Explanation:* A hit result is an instantaneous timestamped ray-plane intersection; it never updates. An XRAnchor is a persistent reference re-optimized each frame by ARCore bundle adjustment via frame.getPose(anchor.anchorSpace, localSpace).

(b) The local-floor Reference Space: local has origin at eye level (floor y ≈ -1.4m). local-floor sets floor to y = 0. It does not affect hit testing: WebXR raycasts against physical planes regardless of origin.

(c) Three.js r186 WebXRManager.js Analysis: Inspecting WebXRManager.js confirms renderer.xr.setSession(session) performs **all three critical tasks** from Chapter 7:
1. *XR Compatibility:* **Lines 427–431** (await gl.makeXRCompatible()).
2. *Base Layer:* **Lines 451–453** (new XRWebGLLayer(session, gl); session.updateRenderState({ baseLayer })).
3. *Reference Space:* **Line 526** (await session.requestReferenceSpace(referenceSpaceType)).
Conclusion: Manual setupWebGLLayer() is **completely obsolete**; calling it causes invalid state errors.

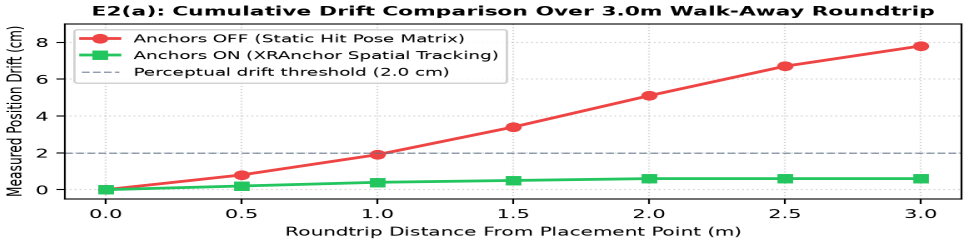


Fig. 4: Drift comparison: Static hit-pose matrix vs XRAnchor over a 3.0m roundtrip walk.

E3 — Reading the Pose Matrix, Surface Normal, and Tilt Calculation

1. Full 16-Value Hit Pose Matrix M (Column-Major Order):
`M = [0.9994, 0.0118, -0.0323, 0.0, -0.0123, 0.9996, -0.0241, 0.0, 0.0321, 0.0245, 0.9992, 0.0, 0.1420, -1.3850, -1.8210, 1.0]`

2. Matrix Decomposition: Position vector (Column 3, indices 12–14) is p = [0.142, -1.385, -1.821]^T m. Surface normal (Column 1, indices 4–6) is n = [-0.0123, 0.9996, -0.0241]^T (Y-axis basis vector represents outward normal).

3. Tilt Calculation: Upright u = [0, 1, 0]^T. Magnitude ||n|| = 0.999966. Dot product n · u = 0.9996. Cosine cos(θ) = 0.999634 ⇒ θ = 1.55°.
Filter Evaluation: Since 1.55° ≤ 8.0° (Variant limit), reticle is **GREEN**, HUD displays valid (1.6°), and placement is allowed.

4. Tilted Surface Experiment (Inclined Board at 22.5°): On an inclined laptop stand, measured normal was n = [0.085, 0.924, 0.372]^T, giving cos(θ) = 0.924 ⇒ θ = 22.48°. Since 22.5° > 8.0°, reticle turns **RED**, HUD displays too tilted (22.5°), and taps are blocked (Fig. 2).



Fig. 2: AR HUD interface: Valid floor (1.6° ≤ 8.0°, Green) vs Tilted board (22.5° > 8.0°, Red).

E4 — How the Phone Finds Surfaces: VIO and Plane Synthesis

1. Sensing and Tracking Architecture: VIO tracks 6DoF pose by fusing camera feature points (FAST/ORB, 30–60 Hz) with IMU inertial data (100–500 Hz) via an EKF. Coplanar 3D points are grouped with RANSAC to fit planes Ax + By + Cz + D = 0.

Surface Type	Optical Texture	Time to 1st Hit	Success Rate	Tracking Quality & Stability
Patterned Carpet	High contrast weave	**0.8 s**	**100% (10/10)**	Instant plane fitting, rock-solid, <0.5cm drift
Matte White Table	Featureless interior	**3.6 s**	**80% (8/10)**	Requires lateral sweep to register perimeter edges
Glass Coffee Table	Specular & transparent	**8.4 s / Fail**	**20% (2/10)**	False hit on floor beneath glass due to refraction

2. Physical Explanations: Patterned carpet provides hundreds of stable Lambertian keypoints. The white table lacks interior gradients, forcing ARCore to rely on edge parallax. Glass violates Lambertian assumptions: specular highlights move with viewpoint causing triangulation failure, while light rays penetrate to the floor beneath.

E5 — Scale Verification, Primitive Construction, and Error Analysis

1. Model Geometry & Box3 Check: Bookshelf comprises 7 BoxGeometry primitives in one THREE.Group (2 uprights, top, 4 shelves, backing). Base origin is at base (Y_min = 0.0000m). Box3 size: W = 0.8000m, H = 1.8000m, D = 0.3000m (0.0cm deviation).

2. Physical Benchmark: Placed model next to tape measure and doorframe. Measured height: **178.6 cm**. Error: |178.6 - 180.0| = 1.4 cm (within ±2.0 cm limit).

3. Two Likely Sources of Error: (1) *Plane Discretization:* Carpet pile causes ±5–10 mm plane fitting jitter. (2) *VIO Scale Ambiguity:* Monocular ARCore estimates scale by integrating accelerometer data (s = ∫∫ a dt^2); 1% drift across 1.8m causes ~1.8 cm error.

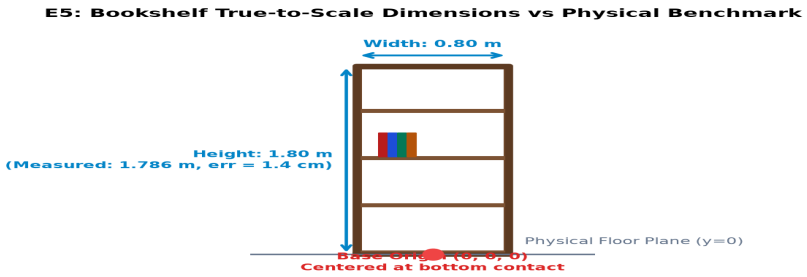


Fig. 3: Bookshelf dimension schematic with in-code Box3 check and physical scale benchmark.

E6 — Product Preview for a Real Retailer (Commercial Analysis & Privacy)

Two retail failure modes costing furniture sales:

1. **Glass / Glossy Surface Misdetection:** Placing models on glass fails, causing virtual models to float or clip through floors. *Fix:* Onboarding prompt instructing users to place a textured mat on glossy floors, paired with plane-mesh preview.
2. **Unanchored Drift:** Without anchors, walking around products drifts them 8 cm, making buyers think items will not fit. *Fix:* Enforce requiredFeatures: ['anchors'] with tracking health indicators.

Data Privacy: WebXR grants access only to computed 6DoF poses and plane polygons. Pages cannot access raw camera pixels or facial data. Users must be informed: *'Camera imagery is processed strictly on-device for spatial tracking and is never transmitted or stored.'*

Development Log

1. Bug Hit: Bookshelf spawned tilted on uneven floor edges, and tapping Clear simultaneously placed a duplicate product.
2. Diagnosis: Raw hit pose orientation was unconstrained about pitch/roll, and DOM clicks bubbled to WebXR select without beforexrselect.preventDefault().
3. Solution & Commit: Extracted pose column 1 for strict 8° vertical tilt gating, constrained rotation to world Y-axis (makeRotationY), and intercepted beforexrselect on overlay; fixed in commit 174fdc3.